

2.5 MW Pulse Power Supply

Team Number: 57

Team Name: *Pulse Power Innovators*

Team Members:

Nicholas Bockrath

Sean Reeves

Diyorbek Egamov

Nicholas Pelino

Project Mentor: Jesus Alejandro Valladares (*Los Alamos National Laboratory*)

Point of Contact: chuy@lanl.gov

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Executive Summary

Group 57, *Pulse Power Innovators*, is developing a low-power, proof-of-concept prototype that demonstrates how pulsed power systems operate in high-energy applications. The project is inspired by the 2.5 MW Pulse Power Supply (Modulator) used at Los Alamos National Laboratory, which delivers powerful, controlled pulses to drive klystrons and other high-voltage equipment.

The goal is not to reproduce the full-scale system but to demonstrate the fundamental operation of a pulse power modulator safely and accessibly. The prototype will generate controlled voltage and current pulses at low power levels (12–24 V) to model how energy is stored and released in larger machines. This project shows that the team understands the principles of pulse formation, energy transfer, and switching control that make high-power modulators possible.

The project's scope includes simulation, design, and construction of a tunable low-voltage pulse generator. The prototype will be tested and evaluated for pulse quality, timing control, and repeatability. This demonstration will provide a clear, educational representation of pulsed power concepts while showcasing the team's design, analysis, and testing abilities.

Although the concept is not unique in the research world, the project's value lies in its educational and practical relevance. The prototype reflects the same physics and engineering that power advanced equipment used in national laboratories, aerospace systems, and medical imaging. By successfully creating and demonstrating this prototype, the team will show that they can understand, analyze, and construct systems inspired by industrial and research-grade pulsed power machines.

The customers and manufacturers of this work include Los Alamos National Laboratory and other organizations involved in high-power, high-voltage systems. These institutions can benefit from the project in two ways. First, by identifying students who have developed hands-on understanding of pulsed power systems, they gain access to a future workforce already familiar with the principles behind large-scale modulators. Second, the prototyping process itself may yield new, innovative approaches to pulse control or energy delivery, which could offer practical improvements to existing systems.

The team will complete simulations, build the hardware prototype, and test the circuit to verify its operation. Results will be documented and presented for review by the project

mentor, Jesus Alejandro Valladares, of Los Alamos National Laboratory.

Introduction and Problem Statement

Pulsed power delivers bursts of energy by storing charge slowly and releasing it very quickly through a controlled switching network. Unlike sinusoidal sources, pulsed systems often resemble square or triangular waveforms in their voltage/current profiles, enabling **high instantaneous power** from moderate average power sources. This operating mode is critical to applications that require short, intense energy delivery: particle accelerators (klystrons, electron guns), radar, certain medical devices, and industrial systems.

Designing, maintaining, and troubleshooting **megawatt-class modulators** is impractical in typical student laboratories due to safety, cost, and facility constraints. Yet these systems underpin technologies used in national security, aerospace, infrastructure, and advanced research. This creates a gap between what students can safely build and what industry actually runs.

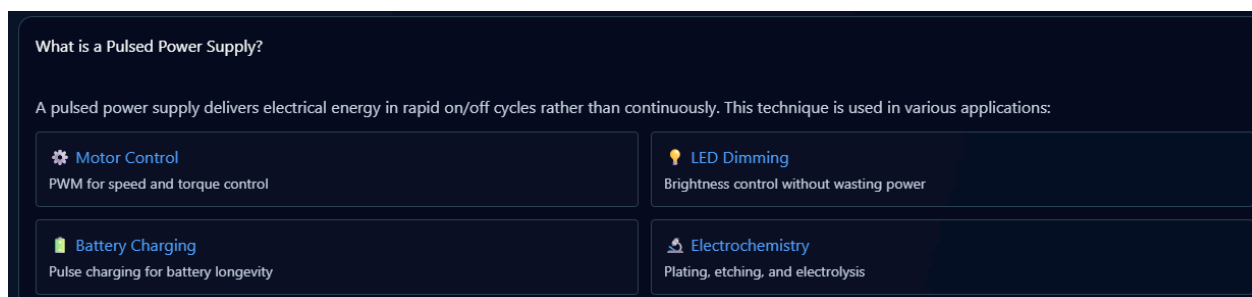


Figure 1: Showcases applications of pulsed power

Problem / Need. The field needs more engineers who understand pulsed-power behavior, safety, and measurement. Every installation is somewhat customized; requirements (voltage, current, pulse width, repetition rate, duty factor) differ across missions and facilities. Because full-scale systems are not standardized and are unsafe for teaching labs, students rarely gain hands-on experience. An **accessible, low-voltage educational prototype** closes this gap by making pulse formation, energy transfer, and waveform control **visible, measurable, and repeatable**.

Educational and Workforce Relevance. For students, the prototype turns theory into practice: it connects circuit models to observable scope traces, and timing diagrams to real switching events. For institutions such as Los Alamos National Laboratory and similar organizations, the project identifies motivated trainees who can transition into

high-reliability environments with a baseline of relevant skills and safety awareness.

What this project demonstrates. The team will demonstrate the internal energy conversion of a modulator: stored energy is shaped into controlled pulses whose **timing, width, repetition rate, and current** are governed by the design, even when operating at safe low voltages. This shows that one can perform “high-energy work” not by maintaining enormous continuous power, but by **storing modest energy and releasing it rapidly**.

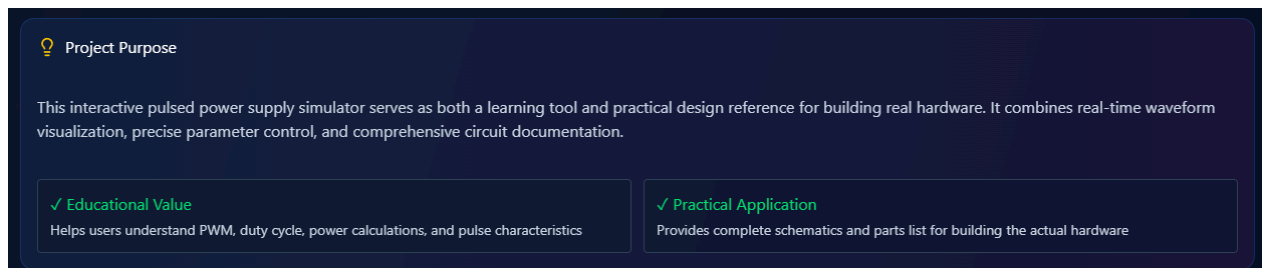


Figure 2: Elaborates on the application of pulsed power

Prototype Application Focus - Medical Pulse Demonstrator

To make the demonstration concrete and medically relevant, the project adopts a **TENS-style constant-current biphasic pulser** as the experimental model. The prototype will:

- Operate from **12–24 V** input for safety and accessibility
- Deliver **charge-balanced biphasic pulses** to a dummy load (e.g., **1 k Ω**) that emulates tissue impedance.
- Support **1–20 mA** output current, **50–500 μ s** pulse width, and **1–100 Hz** repetition rate.
- Provide timing control via a small MCU/timer and current regulation via an op-amp/MOSFET stage.

This choice mirrors real medical pulse delivery (timing accuracy, charge balance, current compliance) while remaining inexpensive and human-safe. It also gives clean oscilloscope waveforms for instruction and reporting.

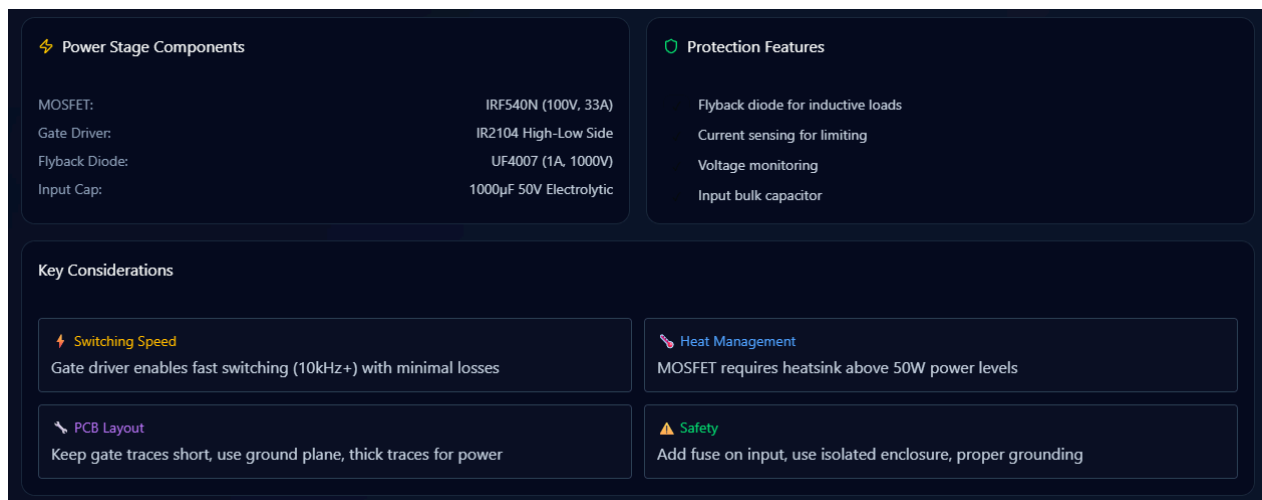


Figure 3: Overview of requirement and important features to consider when designing

Broader Impact. The project strengthens education-to-industry pathways. It provides a tangible platform for students to learn safe handling, measurement, and analysis of pulsed systems; helps mentors evaluate candidates on relevant skills; and may reveal incremental improvements or cost-savvy practices suitable for resource-constrained labs and field settings.

Written Report Roadmap. The remainder of this report is organized as follows. First, the Proof of Concept Approach is presented, outlining how the team will design a low-voltage prototype to model a high-power modulator. Then, Team Capabilities and Facilities is discussed, describing the members' responsibilities and their respective available resources. Afterwards, the Project Scope is laid out, and the Measurements and Success Criteria are explained. Finally, the ABET Considerations are discussed, focusing on sustainability, manufacturability, and safety, and in conclusion, a summary of the expected outcomes and supporting appendices for the budget and schedule is provided.

Customer Discovery

Proximate customer: Project mentor **Jesus A. Valladares** and collaborating faculty.

Ultimate customers: Organizations that design, operate, or maintain high-power pulsed systems (national labs, aerospace/defense contractors, specialized

medical/industrial equipment makers).

Need & Value Proposition. Factories need engineers who can read schematics, simulate switching stages, verify timing, measure pulses, and document results. This project creates a low-risk pathway for training and screening such candidates. It also produces a repeatable demonstrator that can be reused for outreach and on-boarding.

Proof of Concept Approach

The team will build a low-power prototype with a pulse output of about 1 kW in order to verify the selected topology. With laboratory-safe voltage and current levels, this model will mimic the timing and switching properties of a full-scale 2.5 MW modulator. The proof of concept will concentrate on three quantifiable results: **1. Pulse Waveform Fidelity:** Produce 1 ms-wide pulses with steady repetition at 120 Hz and little droop. **2. Switching Synchronization:** Verify the timing control of many solid-state devices to deliver pulses simultaneously. **3. Thermal and Efficiency Metrics:** To evaluate scalability, measure component heating and switching losses under repeated pulsing. The prototype will have a programmed DC power source for consistent operation and a resistive dummy load to mimic klystron impedance.

Team Capabilities and Facilities

Roles

- **Sean Reeves:** Simulation lead, topology selection, parameter sweeps, waveform analysis
- **Nicholas Bockrath:** Data analysis, documentation lead, presentation support, mentor contact
- **Diyorbek Egamov:** Testing lead, hardware assembly, planning
- **Nicholas Pelino:** Hardware assembly lead, testing, planning

Hybrid collaboration and constraints. Three members are remote (Nevada, Midwest, East Coast), two are in-person in Arizona, and the mentor's schedule

varies. To manage this, the team will maintain a weekly cadence with shared notes, version-controlled schematics, and a single **build owner** for the physical prototype. If shipping is required, packaging and post-shipment validation checklists will be used to prevent damage and maintain configuration control.

Facilities & Tools. LTSpice for simulation; access to a basic electronics bench (DC supply, DMM, entry-level oscilloscope); standard hand tools and soldering. Where needed, low-cost USB logic analyzers and current-sense shunts will be employed to capture timing and pulse shape.

Scope

The project will deliver a **low-voltage, low-current prototype** that emulates the operating characteristics of a large modulator while remaining safe and affordable. Scope includes:

- **Simulation:** Topology selection; LTSpice models of the switch stage, current regulation, and timing; waveform validation (pulse width, rise/fall, repetition rate, charge balance).
- **Electronics design:** Op-amp current source with MOSFET regulation; MCU/timer control; over-current and compliance limits; output enable/disable.
- **Prototype build:** Perfboard or simple PCB; labeled test nodes; connectors for dummy loads; EMI-aware wiring and grounding.
- **Measurement & verification:** Oscilloscope captures of current/voltage pulses; measurement of pulse width accuracy ($\pm 5\%$), repetition rate stability, and charge balance for biphasic operation.
- **Safety:** TVS protection, bleeder resistors, insulated connectors, and documented test procedures.

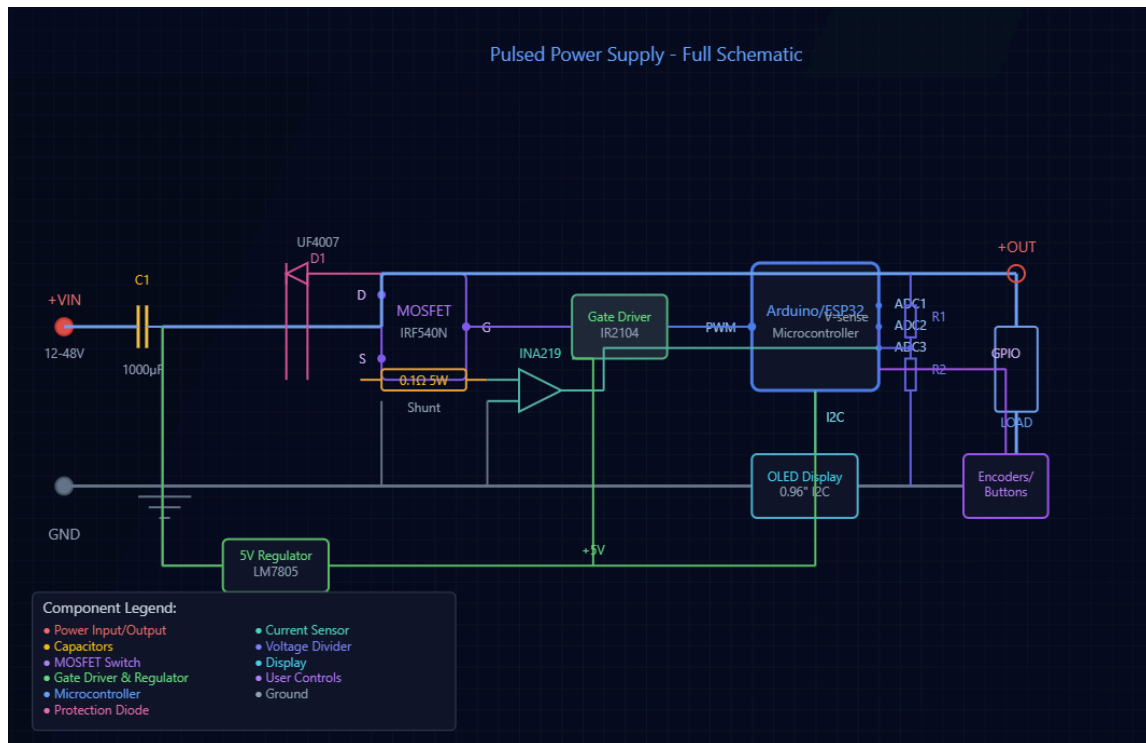


Figure 4: Schematic of entire pulsed power design with components labeled

Milestone	Description	Target Completion
Simulation & Design Selection	LTspice and MATLAB modeling of PFN and switching circuits	Sep 2025
Prototype Development	Construction of low-power IGBT-based pulser	Oct 2025
Proof-of-Concept Testing	Validation of 1 ms pulse generation and timing accuracy	Nov 2025
Performance Analysis	Compare measured vs simulated efficiency and pulse shape	Dec 2025
Final Report & Demo	Summarize findings, propose full-scale design recommendations	Dec 2025

Technical Work Done

Traditional klystron modulators have historically used line-type pulse-forming networks with thyratrons/ignitrons, producing 100-400 us, 100-150 kV pulses at high currents [4]. These systems are proven to be reliable but bulky and limited in pulse flexibility. Modern designs have since shifted toward solid-state solutions using series/parallel-stacked MOSFETs, improving pulse control, efficiency, and reliability [5]. Converter-based and Marx-type architectures extend this approach by charging low-voltage cells and stacking them during the pulse, making digitally controlled flattops and scalable voltage possible [6]. These findings support our team's direction toward a solid-state, Marx-inspired concept and guide how we structure the prototype. Additionally, we reviewed biphasic current-controlled circuits used in biomedical

devices such as TENS systems, which rely on constant-current regulation and charge-balanced pulses [7]. These systems solve similar problems as the klystron modulators, making them a useful reference for pulse regulation and preventing DC offset in the prototype [8].

Industry Landscape. Commercial designs reinforce these trends. ScandiNova's solid-state modulators use modular low-voltage building blocks to achieve high stability and low maintenance [9], while DTI's Marx-based systems demonstrate fast rise times, flat-top regulation, and strong scalability [10]. Legacy PFN/thyratron systems are still being used, but they face higher losses and maintenance burdens [4]. Currently, industry offerings rely on SiC MOSFETs and high-speed IGBTs more and more for improved efficiency and thermal performance [11], [12]. These developments help shape our performance targets and decide on which components to use for the prototype.

Design Process. This semester the team finalized the topology and created the initial design for the prototype. With guidance from our mentor, Jesus Valladares from the Los Alamos National Laboratory, we established the specific parameters for the prototype and outlined the required stages for pulse generation and operation. Early simulations are in progress to confirm switching timing and current and voltage control. These design steps establish the foundation for the next semester.

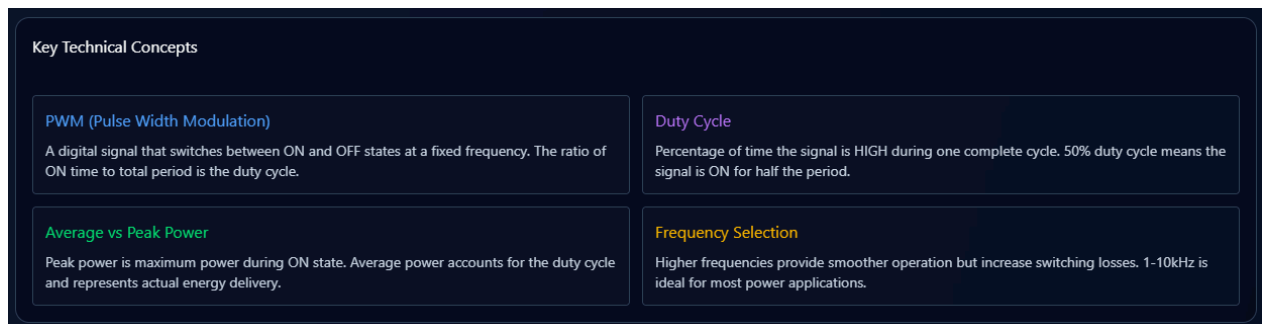


Figure 5: Lists important variables / factors for pulsed power that will be chosen or measured

Technical Work Next Semester

Physical building of the prototype and testing and simulating the build by 2 of the team members in person.

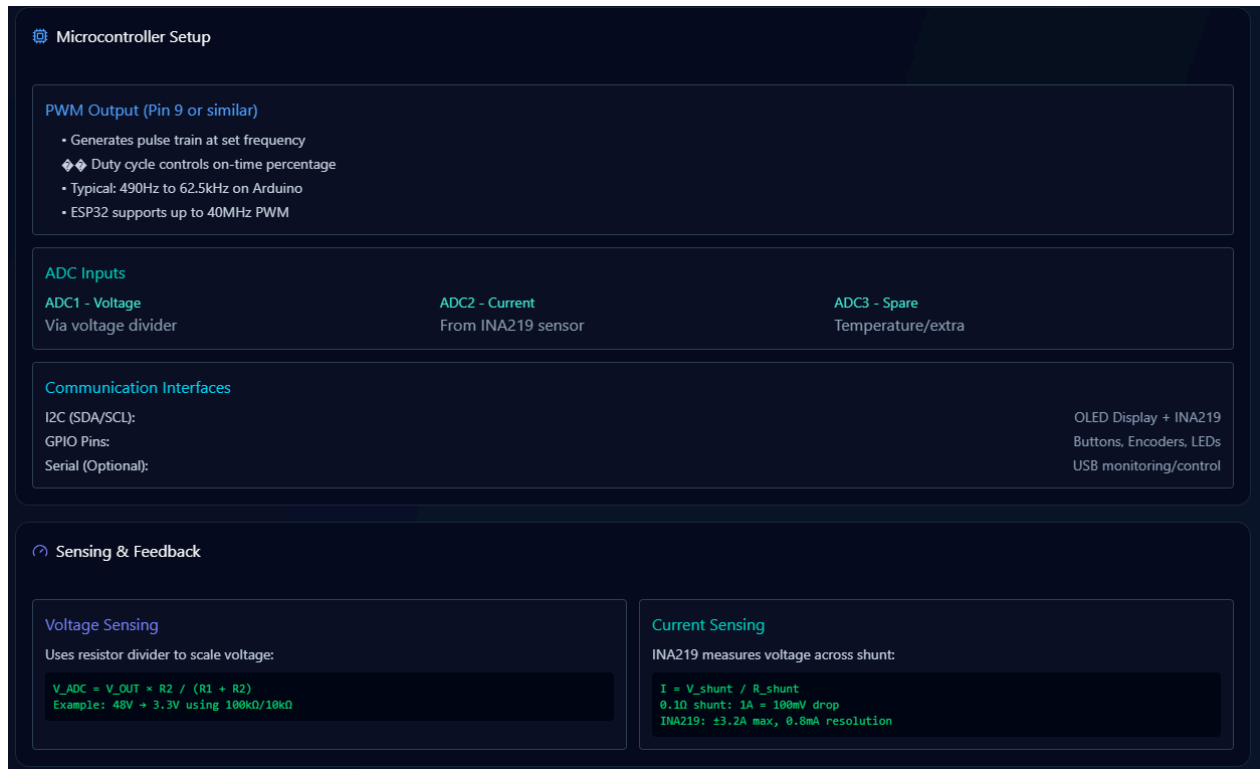


Figure 6: Details the setup required for the microcontroller and circuit

Measurements and Success Criteria

- **Timing:** Pulse width settable **50–500 μs** with $\leq \pm 5\%$ error; **1–100 Hz** PRF with $\leq \pm 2\%$ error.
- **Current regulation:** 1–20 mA across **500 Ω–2 kΩ** loads; flat-top current within $\pm 5\%$.
- **Charge balance:** Integrated biphasic pulse pair net charge $\approx 0 \mu C$.
- **Waveform quality:** Minimal overshoot/ringing with appropriate snubbers; documented rise/fall times.
- **Traceability:** Simulation waveforms correspond to measured lab captures within reasonable tolerance.

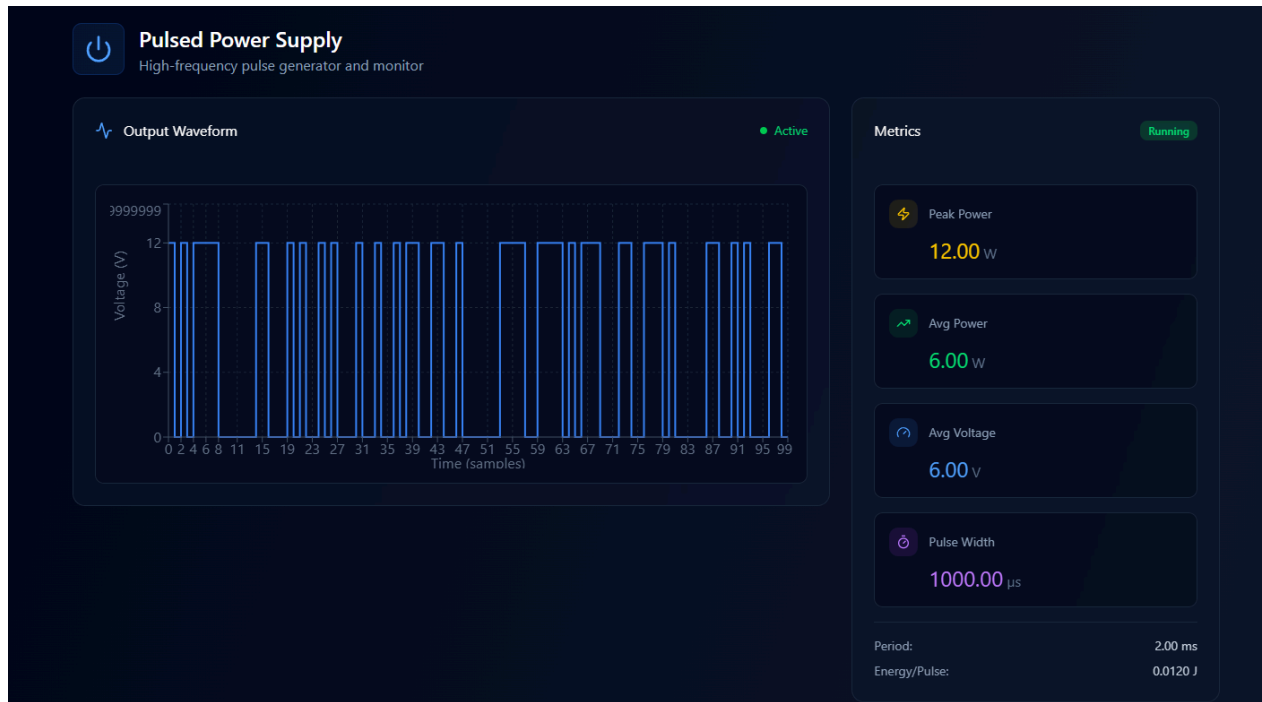


Figure 7: Simulated waveform of a pulsed power generator within the parameters of the project

ABET Considerations

Sustainability: Pulsed power technology is inherently linked to sustainability in the form of its energy usage. The alternative to supplying a quick pulse of voltage is through conventional DC or AC methods, which may not be necessary for all applications. Controlled bursts will **reduce power consumption**, which can be particularly useful in machinery that require extremely high-voltage levels to run, such as a particle accelerator.

Manufacturability: Given the constraints of the supplies allocated to the group, the pulsed power display will be **easier to manufacture** when compared to industrial applications. During the design process, the team is focused on making an affordable demonstration without sacrificing quality of results. This pulse power supply is not meant for high-voltage but to demonstrate to a customer how this method works and how it can be implemented in certain machinery. The team's goal is to design a power supply not only easily comprehensible to a customer, but one that is easily reproducible as well.

Safety: The specific low-voltage topology in this project was chosen with the idea of customer safety in mind. Pulsed power modulation is much more common in high-voltage applications, but due to **safety concerns** commonly present in such situations, a low-voltage model only for demonstration will be used instead. The design will also use devices previously mentioned to ensure safety while testing, such as bleeder resistors and TVS diodes.

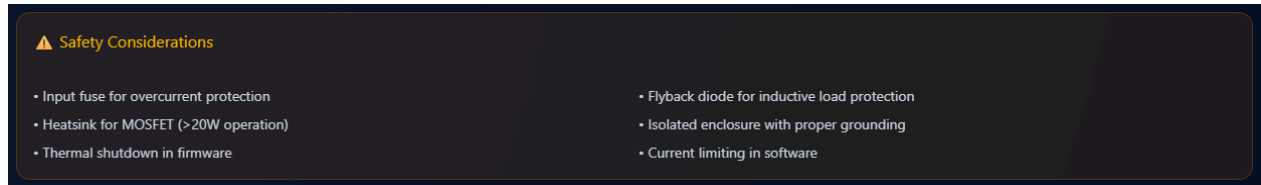


Figure 8: Further safety considerations

Conclusion

This project creates a safe, low-cost platform to **observe and measure pulse formation** with professional discipline. It demonstrates team competency in simulation, design, measurement, and documentation while aligning educational outcomes with the needs of high-power, high-voltage organizations.

Appendix A - Budget

- **Base lab-friendly build:** (Arduino/STM32 or 555; op-amp; 2× MOSFETs; sense resistor; TVS; RC snubbers; enclosure; 12–24 V brick).

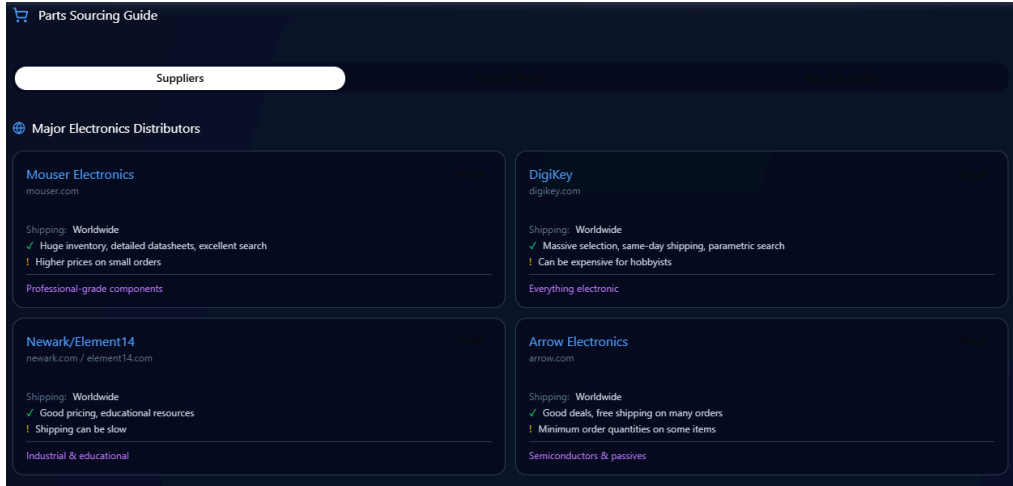


Figure 9: List of distributors for locating materials

- **Notes:** Reuse of existing bench supply/enclosure can reduce cost; add small contingency for shipping or spare parts.

Component	Quantity	Est. Price
Arduino/ESP32	1	\$8-15
MOSFET IRF540N	1-2	\$1-2
Gate Driver IR2104	1	\$2.50
INA219 Module	1	\$5-10
Diode UF4007	2-5	\$1-2
LM7805 Regulator	1	\$0.50-1
OLED Display	1	\$6-10
Capacitors (various)	5-10	\$3-5
Resistors (various)	10-20	\$2-3
Heat sink	1	\$2-5
PCB/Breadboard	1	\$5-15
Wires & connectors	Set	\$5-10
Enclosure	1	\$10-20
Total Estimate:		\$50-120

Figure 11: Complete estimated list of materials with quantity and prices

Appendix B - Schedule

- **Weeks 1–7:** Research, literature review, mentor sync, initial LTSpice environment.

- **Weeks 8–10:** Topology down-select; parameter sweeps; schematic locked for report.
- **Weeks 11–13:** Begin hardware prototyping; bench bring-up; measurement plan.
- **Weeks 14–15:** Capture results; cross-check with simulations; finalize report & presentation.

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